

# Jason (Mukun) Chen

Vancouver, BC, Canada

Email: [mca307@sfu.ca](mailto:mca307@sfu.ca)

Phone: +1-778-238-3427

Personal Blog Info: <https://jcmk77.github.io/>

LinkedIn: [www.linkedin.com/in/jcmk26](https://www.linkedin.com/in/jcmk26)

## Skills

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### Technical Tools:

- **Software:** C/C++, Java, Python, JavaScript, SQL, MATLAB
- **Hardware:** Oscilloscope, VHDL, LTspice, Microcontrollers
- **Platforms:** Linux, Git, Hugging Face, AWS, ROS2, RaspberryPi5
- **Framework:** CPU OpenMP, FPGA HLS, GPU CUDA, RISC-V, LLM, Transformer, CNN, RNN, PyTorch

### Transferable Strength:

- Team Collaboration & Quick Adaptation & Efficient Management & Resilience

## Technical Work Experience

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### ML Engineer Intern — Dr.Peng Group, Beijing, China

Apr–Aug 2023

1. Designed an end-to-end ML pipeline using **Python** and ML libraries, covering data preprocessing and feature engineering, model training and hyperparameter tuning, and performance evaluation to support data analysis.
2. Built pipeline integrations on the cloud server, responsible for data ingestion and processing, persisting results to databases, and supporting testing and iterative validation.
3. Contributed code reviews and data analysis, proposing suggestions aligned with logic needs, promoting collaborative development workflows, and improving cross-team delivery quality and maintainability.
4. Analyzed model outputs with business strategy metrics, produced optimization recommendations, reduced possibility of potential business operation risks, and iterated to improve strategy performance.

### Intern – Embodied AI — CasiaHand Robotics, Beijing, China

May–Dec 2025

1. Built an end-to-end deployment pipeline to run **NVIDIA Isaac GR00T N1.5** (VLA model) on an in-house dexterous-hand platform for grasping objects with **Python**, covering data collection, fine-tuning, inference deployment, and experimental evaluation; improved task success rate from 10% to ~50%.
2. Developed a 2D to 3D data generation AI model application and automated pipeline, including large-scale 2D data collection/cleaning, training/testing, and evaluation; generated 10,000+ 3D asset SKUs.
3. Reproduced multiple open-source DL 3D segmentation models (SAMPart3D / Find3D / SAMesh, etc.), comparing accuracy, speed, and failure modes; delivered feasibility conclusions and improvement recommendations for production-scale deployment.
4. Led the compact development of a dexterous-hand affordance auto-labeling pipeline: performed multi-view sampling for 3D objects and used LLM to generate structured affordance annotations; contributed to a theoretical design of an MLP network for affordance learning with data-labeling.
5. Tracked research papers on embodied AI, multimodal applications, and LLM; provided possible design directions and regular updates on findings.

## Projects

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### High Performance Processor Acceleration in CPU & FPGA & GPU - Heterogeneous Systems

Feb–Apr 2026

1. Accelerated the Rodinia Likelihood Kernel across CPU, GPU, and FPGA platforms in C++, analyzing irregular image-memory access patterns in particle filtering and benchmarking workloads ranging from 100,000 to 3 million particles.
2. Designed an HLS-based FPGA pipeline using tiled load-compute-store buffering, on-chip pixel and likelihood buffers, loop pipelining, array partitioning, AXI transfers, and dataflow buffering to improve throughput and memory efficiency.
3. Implemented and tuned parallel versions with CUDA on GPU, using precomputed mask offsets, simplified likelihood equations, constant-memory storage, and specialized optimization for the fixed 69-point radius-5 mask.
4. Validated aligned, boundary, and stress-test cases in hardware emulation and compared kernel performance across platforms for a 3-million-particle workload.

**DL Language Model on Edge-Device – Deep Learning in Engineering**

Feb–Apr 2026

1. Trained a small language model from scratch for robot-specific system control tasks in **Python** with PyTorch, building the full pipeline from data processing, pre-training, post-training, validation checks, and evaluation.
2. Deployed LoRA fine-tuned open-source language models and custom GGUF models on Raspberry Pi 5 edge devices, benchmarking latency, accuracy, and CPU memory utilization under embedded hardware constraints.
3. Applied quantization and optimization techniques on open-source models (Gemma and Qwen) to reduce model size and improve inference efficiency, balancing accuracy, latency, and hardware utilization cost for practical deployment.
4. Analyzed end-to-end model performance across training and deployment stages, comparing model variants and runtime results to identify the most effective configuration for reliable edge-based robot control.

**AI Agentic Robot Operations Assistant - AI Engineering**

2026 May–In Progress

1. Built a full-stack AI robot operations assistant using Python, API, LangGraph, Streamlit, MCP, ChromaDB, and AI model to support robot status queries, troubleshooting, info lookup, and navigation request management.
2. Designed a LangGraph multi-agent workflow that routes natural language requests to RAG, tool, clarification, or fallback agents based on structured intent classification, entity extraction, and validation results.
3. Implemented LLM and hybrid semantic understanding modes to convert user commands into safe robot operation intents, validation, request IDs, and lifecycle transitions before tool execution.
4. Developed an intuitive user interface and clear project documentation to improve accessibility for end users and simplify ongoing engineering maintenance.

**Autonomous Delivery Robot Development – Engineering Development in Robotics**

2026 Jan–In Progress

1. Developed the navigation and control system for an autonomous transport robot using ROS 2, Nav2, and LiDAR-based SLAM on Raspberry Pi, enabling supervised point-to-point missions between destinations.
2. Integrated map loading, AMCL localization, obstacle detection, costmap-based planning, and differential-drive motion control with encoder feedback to support reliable route execution, obstacle avoidance, and safe-stop behavior.
3. Designed the mission-control layer by implementing destination validation, mission request handling, and a state workflow with IDLE, EN\_ROUTE, PAUSED, and COMPLETED states.
4. Collaborated in a multidisciplinary 4 people team across navigation, mission control, mechanical, and power subsystems, coordinating interfaces, integration, documentation and testing.

**2D Arcade Game Development - Software Engineering**

Jan–Mar 2025

1. Developed a 2D arcade-style maze game in **Java**, using advanced object-oriented design principles to build a modular and extensible software architecture that supports robust gameplay and an engaging experience.
2. Conducted debugging and testing, identifying and resolving critical software defects to optimize performance, reinforce stability, and enhance user satisfaction.
3. Utilized modeling techniques like UML to define class hierarchies, system behaviors, and data structures, ensuring a scalable and maintainable foundation conducive to future expansions.
4. Collaborated within a small agile team, employing **Git** for version control and continuous integration, facilitating efficient task management, code reviews, and iterative development cycles.

**Education****Simon Fraser University** BSc. Computer Engineering

September 2022 - In Progress

Areas of Focus: AI/DL, Hardware Acceleration, Embedded Systems

**Interests**

Part-Time Tour Guide: Guided small tour groups in English or Mandarin in Beijing, China

Swimming Club: Participated championships in Ontario, Canada

Independent &amp; Technical Learning: ML/DL/RL, AI Agent, LLM, AI Infrastructure